What about E = mc
2?

Although Einstein did not advocate calling his theory “relativity” (suggesting instead the name “invariance” theory to reflect the unchanging character of the speed of light, among other things), the meaning of the term is now clear. Einstein’s work showed that concepts such as space and time, which had previously seemed to be separate and absolute, are actually interwoven and relative. Einstein went on to show that other physical properties of the world are unexpectedly interwoven as well. His most famous equation provides one of the most important examples. In it, Einstein asserted that the energy (E) of an object and its mass (m) are not independent concepts; we can determine the energy from knowledge of the mass (by multiplying the latter twice by the speed of light, c

2) or we can determine the mass from knowledge of the energy (by dividing the latter twice by the speed of light). In other words, energy and mass—like dollars and francs—are convertible currencies. Unlike money, however, the exchange rate given by two factors of the speed of light is always and forever fixed. Since this exchange-rate factor is so large (c

2 is a big number), a little mass goes an extremely long way in producing energy. The world grasped the devastating destructive power arising from the conversion of less than 1 percent of two pounds of uranium into energy at Hiroshima; one day, through fusion power plants, we may productively use Einstein’s formula to meet the energy demands of the whole world with our endless supply of seawater.

From the viewpoint of the concepts we have emphasized in this chapter, Einstein’s equation gives us the most concrete explanation for the central fact that nothing can travel faster than light speed. You may have wondered, for instance, why we can’t take some object, a muon say, that an accelerator has boosted up to 667 million miles per hour—99.5 percent of light speed—and “push it a bit harder,” getting it to 99.9 percent of light speed, and then “really push it harder” impelling it to cross the light-speed barrier. Einstein’s formula explains why such efforts will never succeed. The faster something moves the more energy it has and from Einstein’s formula we see that the more energy something has the more massive it becomes. Muons traveling at 99.9